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Physics 135A
Midterm 2

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① Unit 4: Rotational Kinematics

- ① 1.33 rp-s, rpm = ?, angular velocity rad/s = ?

$$F = \frac{1}{1.33} = 0.752 \text{ per sec.}$$

$$\text{rpm} = 0.752 \times 60 \text{ s} = \boxed{45.11 \text{ rpm}}$$

$$1 \text{ rotation} = 2\pi \text{ rad.}$$

$$\omega = 0.752 \times 2\pi = \boxed{4.72 \text{ rad/s}}$$

Answer: B,
45 rpm, 4.7
rad sec⁻¹

- ② centripetal accel? coin on edge of record.

$$a_c = \omega^2 r$$

$$\omega = 4.7 \text{ rad/s}$$

radius of records? → standard
diameter = 12 in or ~30.48 cm
so, $r = 15.24 \text{ cm}$

$$a_c = (4.7)^2 (15.24)$$

$$a_c = 3.36 \text{ m/s}^2 \rightarrow \text{doesn't match answers}$$

if I instead assume the record is 10 cm

$$a_c = \omega^2 r$$

$$a_c = (4.7)^2 (10) \text{ cm}$$

$$a_c = 220.9 \text{ cm/s}^2 = \boxed{2.209 \text{ m/s}^2, \text{ Answer} = D}$$

- ③ blood centrifuge @ angular velocity. $\times 100$, how?

$$a_c = \omega^2 r$$

$$\omega \rightarrow 100 \times$$

$$\sqrt{\omega_2 r} = \sqrt{100 \times (\omega_1)^2 r}$$

$$\omega_2 = 10 \times \omega_1$$

$$\boxed{\text{Answer} = D, \omega \rightarrow 10\omega}$$

- ④ $\omega = 200 \text{ rpm}$, $r = 12 \text{ cm}$, solid mass = 10 g, centripetal force = ?

$$F_c = m \cdot a_c = 0.12 \text{ m}$$

$$F_c = m(\omega^2 r)$$

$$F_c = 0.01 \text{ kg} \times (20.94^2 \times 0.12 \text{ m})$$

$$F_c = 0.01 (52.63)$$

$$F_c = 0.5263 \text{ N, B}$$

$$\boxed{\text{Answer} \approx 0.53 \text{ N}}$$

$$\omega = 200 \text{ rpm} \times \frac{2\pi}{60}$$

$$\omega = 20.94 \text{ rad/s}$$

⑤ 1ax: angular vel. = 30 rad/s , ball = 1.4 m from elbow. $v = ?$

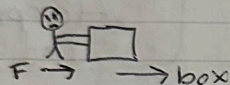
$$v = \omega r$$

$$v = (30 \text{ rad/s}) \times (1.4 \text{ m})$$

$$v = 42.0 \text{ m/s, Answer = C}$$

2. Unit 5: Work & Energy

①



$$\vec{F} = 40\hat{i} - 30\hat{j} \text{ N}$$

$$\Delta \vec{x} = 5\hat{i} \text{ m}$$

Work = ? & btwn force & displacement

$$W = F \Delta x$$

$$W = ((40\hat{i}) - 30\hat{j}) \cdot (5\hat{i})$$

$$W = (40\hat{i})(5\hat{i}) + (-30\hat{j})(0\hat{j})$$

$$W = 200 \text{ Nm}$$

$$\cos \theta = \frac{F \Delta x}{|F| |\Delta x|}$$

$$= \frac{200}{50 \times 5} = \frac{200}{250}$$

$$F = \sqrt{40^2 + 30^2} = \sqrt{2500} = 50 \text{ N}$$

$$\Delta x = 5 \text{ m}$$

$$\cos \theta = 0.8$$

$$\theta = \cos^{-1}(0.8)$$

$$\theta \approx 36.87^\circ$$

Answer: D,
 $W = 200 \text{ Nm}, \theta \approx 37^\circ$

② ball = 0.15 kg

a) $F = 1000 \text{ N}$, displacement = 10 cm , $W = ?$

$$W = F \cdot d \cdot \cos \theta$$

$$W = 1000 \text{ N} \cdot 0.1 \text{ m} \cdot 1$$

$$W = 100 \text{ Nm} = 100 \text{ J}$$

$$W = 100 \text{ J}$$

$$\cos 0^\circ = 1$$

b) $\Delta K = ?$, baseball initially stationary.

$$\Delta K = W = 100 \text{ J}$$

← since stationary

$$\Delta K = 100 \text{ J}$$

part c on next pg.

③ $V_f = ?$

$$K_f = \frac{1}{2} m V_f^2 \rightarrow V_f = \sqrt{\frac{2K_f}{m}}$$

$m = .15 \text{ kg}$
 $K_f = 100 \text{ J}$

$$V_f = \sqrt{\frac{2K_f}{m}}$$

$$V_f = \sqrt{\frac{2 \times 100}{0.15 \text{ kg}}} = \sqrt{\frac{2 \times 100}{0.15}} = \sqrt{\frac{200}{0.15}} = \sqrt{1333.3} \approx 36.5 \text{ m/s} = V_f$$

④ $R = \frac{V_f^2 \sin(2\theta)}{g}$

$V_f^2 = 36.5 \text{ m/s}$
 $\theta = 45^\circ$

$R = \frac{36.5 \sin(2 \cdot 45)}{9.81}$

$g = 9.81 \text{ m/s}^2$

$$R = \frac{36.5(1)}{9.81} = \frac{1332.25}{9.81} = 135.5 \text{ m}$$

③ 40 kg box $\mu_k = 0.05$ for 40 m

a) $W = ?$

$F_{\text{friction}} = \mu_k \cdot N$

$F_f = 0.05 \cdot (40 \cdot 9.81)$

$= 0.05 \cdot 392.4 \text{ N}$

$F_{\text{frict}} = 19.62 \text{ N}$

$W = 19.62 \text{ N} \times 40 \text{ m}$

$W = 784.8 \text{ J}$

b) Power = ?

$P = \frac{W}{t} = \frac{784.8 \text{ J}}{30 \text{ s}} = P = 26.16 \text{ Watts}$

④ 3 W for 1 year is 9 cents/kWh?

$3 \text{ W} \rightarrow 0.003 \text{ kW}$

$1 \text{ yr} = 24 \times 365 = 8760 \text{ hrs}$

Energy = $P \times t$

$0.003 \times 8760 \text{ hr} = 26.28 \text{ kWh}$

$9 \text{ cents} \times 26.28 = \$2.3652 \approx \$2.37 \approx \$2.33, D$

3. Unit 6: Linear Momentum

- ① 2 molecules $\rightarrow$ 1 molecule
each weighs $20 \times 10^{-25} \text{ kg}$. One has $v = 350 \text{ m/s}$, one $v = -350 \text{ m/s}$
final speed (v_f) = ?

$$mv_1 + mv_2 = 2mv_f$$

$$m(350) + m(-350) = 2mv_f$$

$$m(350 - 350) = 2mv_f$$

$$0(m) = 2mv_f$$

$$v_f = 0 \text{ m/s, C}$$

- ② 2 molecules approach w/ momentum vectors 45° apart. $v_f = ?$
- they stick together.

$$v_1 = 350 \text{ m/s}, v_2 = -350 \text{ m/s}$$

$$p_{1x} = m \cdot 350 \cdot \cos(45^\circ)$$

$$p_{1y} = m \cdot 350 \cdot \sin(45^\circ)$$

$$v_{1x} = 350 \text{ m/s}$$

$$v_{1y} = 0 \text{ m/s}$$

$$p_{2x} = m \cdot (-350) \cdot \cos(45^\circ)$$

$$p_{2y} = m \cdot (-350) \cdot \sin(45^\circ)$$

$$v_{2x} = -350 \cos 45^\circ = -350 \left(\frac{\sqrt{2}}{2}\right) = -247.45 \text{ m/s}$$

$$v_{2y} = -350 \sin 45^\circ = -247.45 \text{ m/s}$$

Momentum

$$p_{xt} = m \cdot v_{1x} + m \cdot v_{2x}$$

$$= 20 \times 10^{-25} (0) + 20 \times 10^{-25} (-247.45)$$

$$= -4.95 \times 10^{-23} \text{ kg m/s}$$

$$p_{yt} = m \cdot v_{1y} + m \cdot v_{2y}$$

$$= 20 \times 10^{-25} (350) + 20 \times 10^{-25} (-247.45)$$

$$= 2.05 \times 10^{-23} \text{ kg m/s}$$

velocity

$$p_{yt} = 2m \cdot v_{fy}$$

$$2 \cdot 20 \times 10^{-25} (v_{fy}) = -4.95 \times 10^{-23}$$

$$v_{fy} = \frac{-4.95 \times 10^{-23}}{40 \times 10^{-25}}$$

$$v_{fy} = -12.375 \text{ m/s}$$

$$p_{xt} = 2m \cdot v_{fx}$$

$$2 \cdot 20 \times 10^{-25} (v_{fx}) = 2.05 \times 10^{-23}$$

$$v_{fx} = \frac{2.05 \times 10^{-23}}{40 \times 10^{-25}}$$

$$v_{fx} = 5.125 \text{ m/s}$$

$$v_f = \sqrt{v_{fx}^2 + v_{fy}^2}$$

$$v_f = \sqrt{(5.125)^2 + (12.375)^2}$$

$$v_f = \sqrt{26.266 + 152.89}$$

$$v_f = \sqrt{179.156} = v_f = 13.4 \text{ m/s}$$

- ③ $F = 4000\text{ N}$ on car for 0.20 s
 impulse = ? final velocity = ? $v_i = 2.8\text{ m/s}$ $m = 200\text{ kg}$ ^{cart driver} ~~neglect friction.~~

① $I = F \cdot \Delta t$
 $I = 4000\text{ N} \times 0.2\text{ s}$
 $I = 800\text{ N}\cdot\text{s}$

② $I = \Delta p = m \cdot \Delta v$
 $I = 200 \cdot (v_f - 2.8\text{ m/s})$
 $I = 800\text{ N}\cdot\text{s}$
 $800\text{ N}\cdot\text{s} = \frac{200}{200} (v_f - 2.8\text{ m/s})$
 $v_f = 4.0\text{ m/s} + 2.8\text{ m/s}$
 $v_f = 6.8\text{ m/s}$

- ④ no motion $\rightarrow$ collision.
 same v after collision. original ball stops. this is?
 • total KE is =, conserves momentum & KE. Not totally elastic bc balls will deform.
C. elastic collision.

- ⑤ $30,000\text{ kg}$ @ 0.85 m/s , add $110,000\text{ kg}$

① $v_f = ?$
 momentum_i = $m_{\text{car}} \cdot v_{\text{car}} + m_{\text{metal}} \cdot v_{\text{metal}}$
 $= 30000 \times 0.85 + 110000 \cdot 0$
 $= 25,500\text{ kg}\cdot\text{m/s}$
 momentum_f = $(m_{\text{car}} + m_{\text{metal}}) \cdot v_f$
 momentum conserved, $\rightarrow m_i = m_f$

$25,500\text{ kg}\cdot\text{m/s} = (30,000 + 110,000\text{ kg}) v_f$
 $25500 = 140,000 \cdot v_f$
 $v_f = \frac{25,500}{140,000} = 0.182\text{ m/s}$

- ② on next pg $\rightarrow$

$$\textcircled{b} \quad KE_i = \frac{1}{2} m_{\text{car}} v_{\text{car}}^2 + \frac{1}{2} m_{\text{metal}} v_{\text{metal}}^2$$

$$KE_i = \frac{1}{2} (30000) (0.85)^2 + \frac{1}{2} (110000) (0)^2$$

$$KE_i = 10,837.5 \text{ J}$$

$$KE_f = \frac{1}{2} (m_{\text{car}} + m_{\text{metal}}) v_f^2$$

$$= \frac{1}{2} (30000 + 110000) (0.82)^2$$

$$KE_f = 2,316.66 \text{ J}$$

$$KE_{\text{lost}} = KE_i - KE_f$$

$$= 10,837.5 \text{ J} - 2,316.66 \text{ J}$$

$$KE_{\text{lost}} = 8,518.82 \text{ J}$$

4. Unit 7: Statics

- ① Steel beam
hinged
- 40 kg 3m from origin
 - 60 kg 2m from origin.

$$T = F \cdot d$$

$$W = m \cdot g$$

$$W_{\text{brick}} = m \cdot g$$

$$= 40 \cdot 9.81$$

$$W_{\text{brick}} = 392 \text{ N}$$

$$T_{\text{brick}} = W \cdot d$$

$$T = 392 \cdot 3 \text{ m}$$

$$T_b = 1176 \text{ N} \cdot \text{m}$$

$$W_{\text{concrete}} = m \cdot g$$

$$= 60 \cdot 9.81$$

$$W_{\text{conc}} = 588 \text{ N}$$

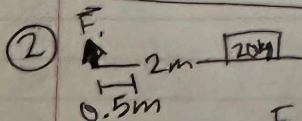
$$T_{\text{conc}} = W \cdot d$$

$$T_{\text{conc}} = 588 \text{ N} \cdot 2 \text{ m}$$

$$T_c = 1176 \text{ N} \cdot \text{m}$$

same, no motion.

A. remains motionless.



net force & torque = 0

$$F_{\text{gravity}} = m \cdot g$$

$$= 20 \text{ kg} \times 9.81 \text{ m/s}^2$$

$$F_{\text{gravity}} = 196 \text{ N} = F_2 - F_1$$

$$F_2 - 98 \text{ N} = 196 \text{ N}$$

$$F_2 = 196 + 98$$

$$F_2 = 294 \text{ N}$$

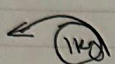
$$T_1 = T_{\text{gravity}}$$

$$F_1 \cdot 2 \text{ m} = 196 \text{ N} \cdot 1 \text{ m}$$

$$F_1 = \frac{196}{2} = 98.0 \text{ N}$$

$$F_1 = 98 \text{ N}, F_2 = 294 \text{ N}$$

5. Unit 8: Rotational Dynamics

①  $I = \frac{1}{2} MR^2$

② gear is 1 kg, $r = 8 \text{ cm}$, I in $\text{kg} \cdot \text{m}^2$?

$$I = \frac{1}{2} MR^2$$

$$I = \frac{1}{2} (1 \text{ kg}) (0.08 \text{ m})^2$$

$$I = 0.0032 \text{ kg} \cdot \text{m}^2$$

③ $T = ?$ $\alpha = 2 \text{ rad/s}^2$ to spin the gear.

$$T = I \cdot \alpha$$

$$T = (0.0032 \text{ kg} \cdot \text{m}^2) (2 \text{ rad/s}^2)$$

$$T = 0.0064 \text{ N} \cdot \text{m}$$

②



$$\vec{r} = -20\hat{i} + 20\hat{j} \text{ cm}, \vec{\tau} = \vec{r} \times \vec{F}$$

③ $\vec{F} = 12\hat{i} \text{ N}$, $T = ?$

$$\vec{\tau} = \vec{r} \times \vec{F}$$

$$\vec{\tau} = (-20\hat{i} + 20\hat{j}) \text{ m} (12\hat{i} \text{ N})$$

$$\text{Cross product: } \tau_k = (r_x F_y - r_y F_x)$$

$$\tau_k = (-.2)(0) - (.2)(12)$$

$$\tau_k = -2.4 \text{ N} \cdot \text{m}$$

④ force vector?

$$\tau = -2.4\hat{k} \rightarrow T = 2.4\hat{k} \text{ N} \cdot \text{m}$$

$$\vec{\tau} = \vec{r} \times \vec{F}$$

$$\tau_k = r_x F'_y - r_y F'_x$$

$$2.4 = (-.2)(F'_y) - (.2)(F'_x)$$

$$2.4 = -.2F'_y - .2F'_x$$

$$-12 = F'_y - F'_x$$

We know $F'_x = 0$, so $F'_y = 12$

$$\vec{F}' = 0\hat{i} - 12\hat{j} \text{ N}$$

$$F' = -12\hat{j} \text{ N}$$